

Math 421

Monday, November 10

Announcements

- Exam Week Drop-In Hours
 - Tuesday 1:30-3:30 pm VV B205
 - Wednesday 12:00-1:00 pm VV B205
- Exam 2
 - **Date:** Wednesday, 11/12
 - **Time:** 5:45-7:15 pm
 - **Location:** Birge 145

Continuity – Equivalent Definitions

Theorem. Let $f: A \rightarrow \mathbb{R}$ and let $x_0 \in A$. Then the following are equivalent:

- a) f is continuous at x_0 ,
- b) **via sequences:** for every sequence (x_n) , $x_n \in A$, if $\lim x_n = x_0$ then $\lim f(x_n) = f(x_0)$.
- c) **via $\epsilon - \delta$:** for $\epsilon > 0$, there exists $\delta > 0$ such that $|x - x_0| < \delta$ implies $|f(x) - f(x_0)| < \epsilon$.
- d) $\lim_{x \rightarrow x_0} f(x) = f(x_0)$.

Activity – Prove Using $\epsilon - \delta$ Definition

1. If $a \in \mathbb{R}$ and $g(x) = a$ is the constant function, then g is continuous at all $c \in \mathbb{R}$.

Let $\epsilon > 0$. Set $\delta = \underline{\hspace{2cm}}$. Assume $|x - c| < \delta$, then

$$|g(x) - g(c)| = \underline{|a - a| = 0 < \epsilon}.$$

Since $\epsilon > 0$ was arbitrary, $g(x)$ is continuous at $c \in \mathbb{R}$.

2. If $f(x) = x$, then f is continuous at all $c \in \mathbb{R}$.

Let $\epsilon > 0$. Set $\delta = \underline{\epsilon}$. Assume $|x - c| < \delta$, then

$$|f(x) - f(c)| = \underline{|x - c| < \delta = \epsilon}.$$

Since $\epsilon > 0$ was arbitrary, $f(x)$ is continuous at $c \in \mathbb{R}$.

Algebraic Continuity Properties

Theorem 17.3 & 17.4. Assume $f: A \rightarrow \mathbb{R}$ and $g: A \rightarrow \mathbb{R}$ are both continuous at a point $x_0 \in A$. Then

1. $|f(x)|$ is continuous at x_0 ,
2. $kf(x)$ is continuous at x_0 for all $k \in \mathbb{R}$,
 $\lim x_n = x_0$ then $\lim kf(x_n) = kf(x_0)$.
3. $f(x) + g(x)$ is continuous at x_0 ,
4. $f(x)g(x)$ is continuous at x_0 , and
5. If $g(x_0) \neq 0$, then $\frac{f(x)}{g(x)}$ is continuous at x_0 .

Proof of 2. Let (x_n) be a sequence, $x_n \in A$, such that $\lim x_n = x_0$. Since f is continuous at x_0 , $\lim f(x_n) = f(x_0)$. So $\lim kf(x_n) =$

$$k \cdot \lim f(x_n) = kf(x_0).$$

By our algebraic sequence limit properties.

Example

$$P(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n \quad n \in \mathbb{N}$$
$$a_0, a_1, \dots, a_n \in \mathbb{R}.$$

1. If P is a polynomial, then P is continuous at all $c \in \mathbb{R}$
2. Define $f: [0, \infty) \rightarrow [0, \infty)$ by $f(x) = \sqrt{x}$. Then f is continuous at all $c \in [0, \infty)$. let (x_n) be a sequence, $x_n \in [0, \infty)$, such that $\lim x_n = c$. Then (proved on 10/01 or in §8)
 $\lim \sqrt{x_n} = \sqrt{c}$. Since (x_n) was an arbitrary sequence,
 $f(x) = \sqrt{x}$ is continuous at $c \in [0, \infty)$.
3. $\sin(x)$, $\cos(x)$, e^x , $\ln(x)$ - continuous on their domains.

$$f: (-\infty, 0)$$

$$f(x) = x^2$$

$$g = \sqrt{x}$$

$$(g \circ f)(x) = \sqrt{x^2}$$

$$f(x) = \sqrt{1+x}$$

$$g(x) = \frac{1}{x}$$

$$(g \circ f)(x) = \frac{1}{\sqrt{1+x}}$$

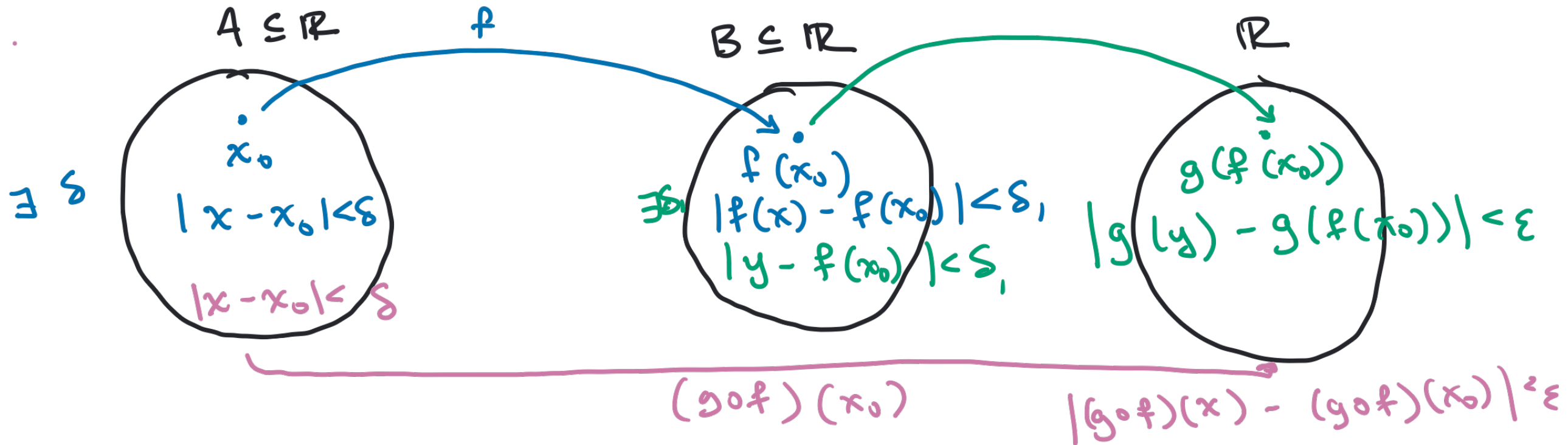
Composition of Continuous Functions (-1, \infty)

Theorem 17.5. Suppose $f: A \rightarrow \mathbb{R}$ and $g: B \rightarrow \mathbb{R}$. Assume

$\{f(x): x \in A\} \subseteq B$ so that the composition $(g \circ f)(x) = g(f(x))$ is defined on A .

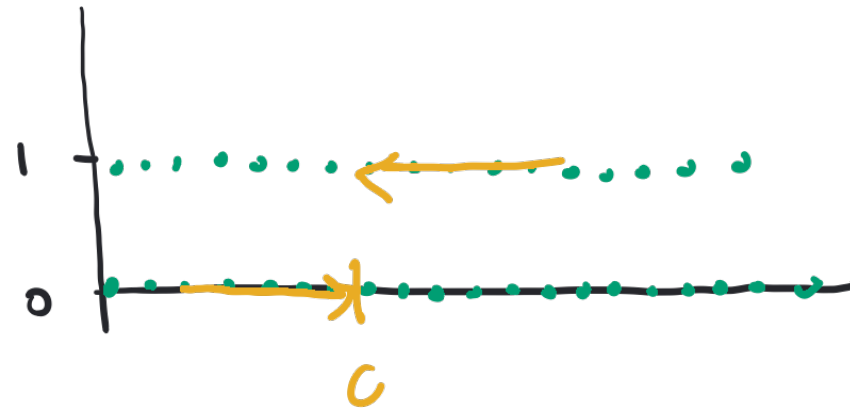
If f is continuous at $x_0 \in A$ and g is continuous at $f(x_0) \in B$, then the $g \circ f$ is continuous at x_0 .

Let $\varepsilon > 0$.



Example

$$\text{Let } f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$$



Fact: $f(x)$ is not continuous at any point $c \in \mathbb{R}$.